3. EEG has proved to be a useful diagnostic tool in cases of serious head injuries, brain tumours and cerebral infections.
4. It also helps to find the diseases like epilepsy and various degenerative disease of the nervous system.
5. EEG is useful in assessing patients with suspected brain death.

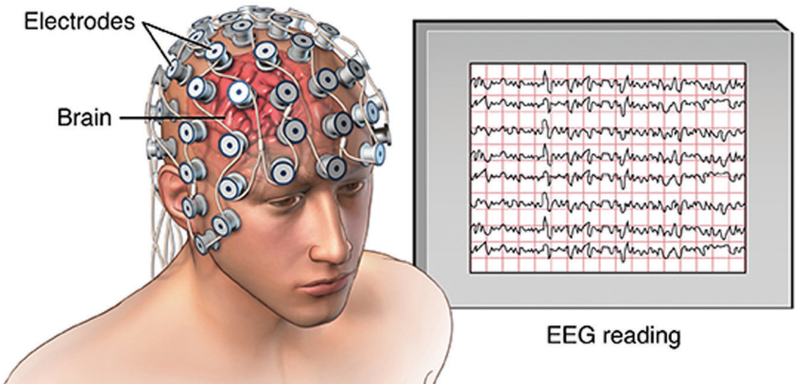


Figure 12.5 EEG with waves

Wearable Healthcare devices include smart watches, fitbit wristbands, fitness trackers, helmet-worn impact checkers, back pain posture checkers, necklaces, clip-ons, smart clothing and filament implant. These medical devices with select sensors can record real-time data on various physiological conditions and electronically send them to doctor, or via apps on smart phone. It has two main fitness apps such as **Activity app and Work out app.** Activity app is all about health, movement, wellness and daily routine; and **Workout app** tracks running, cycling and walking. The wearable healthcare devices are activity trackers, which offer health benefits by measuring number of steps, distance travelled, calories burned, heart rate, detects blood volume and capillary-size changes under pressure, body temperature and quality of sleep. It also tracks progress over time so that one can set daily goals for bodyweight, blood glucose level, blood pressure, sleep time, water intake, etc.

X-rays

Radiography is the use of X-rays to visualize the internal structures of a patient (Figure12.6). X-Rays are a form of electromagnetic radiation, produced by an X-ray tube. The X-rays are passed through the body and captured behind the patient by a detector; film sensitive to x-rays or a digital detector. There is variation in absorption of the X-rays by different tissues within the body. Dense bones absorb more radiation, while soft tissue allows more rays to pass through. This variation produces contrast image within the image to give a 2D representation of all the structures. It is invasive. It is also available as a portable X ray unit. It is less costly when compared to the other imaging units like MRI or CT.

Clinical significance

1. X- ray imaging is used for diagnosing the disease of the heart, lungs and fractures of bones and joints
2. It is also used to visualise hollow organs and blood vessels by filling them with certain chemical formulations containing barium and iodine
3. Dental radiography is used in diagnosis of oral problems
4. Mammography is a special type of X-ray imaging to create detail images of the breast tissues
5. Fluoroscopy for real time images
6. X-rays are used in radiation therapy to shrink cancerous tumours

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